

Allen Abraham

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Professional Summary

Software Engineer specializing in cloud pipelines (**AWS** and **GCP**) and **distributed systems**. Mid-level professional delivering **production-ready** solutions, including an **Airflow DAG** that cut data ingestion time by 99.8% and a refunds platform handling over 1 million transactions per hour with **99.999% uptime**. Experienced in building scalable **microservices**, ETL workflows, and real-time monitoring dashboards. Looking to leverage my expertise to drive **reliable, high-performance** infrastructure in a fast paced environment.

Education

Master of Science in Computer Science | *New York University*

08/2023 - 05/2025

GPA: **3.96** / 4.0

Bachelor of Engineering in Computer Science | *Visvesvaraya Technological University*

08/2017 - 05/2021

GPA: **8.33** / 10.0

Work Experience

Canonical (Ubuntu) | *Software Engineer I*

08/2025 - Present

- Driving the AI initiative in the CPC team to utilize AI tools in the workspace and develop an agentic tool to automate documentation tasks using a Python pipeline.
- Modernizing the Ubuntu build tool using imagecraft in **Jenkins**, leading to performant and maintainable tooling for latest Ubuntu builds.
- Improving **integration tests** of builder images to catch bugs and misses in our pipelines and to produce more stable Ubuntu images

Firebird Music | *Data Engineer Intern*

06/2024 - 08/2024

- Developed an **Airflow DAG** on Cloud Composer to incrementally load data from an S3 bucket into partitioned **BigQuery** tables via Cloud Storage, reducing ingestion time by **99.8%** and saving **\$10K** annually by eliminating full table refreshes.
- Collaborated with a globally distributed team to analyze and interpret social media images for over 50,000 artists using the **GPT-Vision** API, uncovering engagement trends that informed marketing strategy and guided artists on image-driven improvements.

Juspay | *Software Engineer*

11/2020 - 07/2023

- Optimized gateway scoring service by implementing **Groovy-based algorithms**, increasing customer success rates by **up to 30%**.
- Owned the refunds vertical, introducing features such as a **PostgreSQL** and Redis-based distributed producer-consumer workflow and scalable, **REST APIs** - scaling to process over **1 million refunds** per hour with a median transaction amount of \$100.
- Built Prometheus and ClickHouse Grafana dashboards on GCP for **real-time metric** analysis, maintaining 98% refund success & **99.999% uptime**, slashing debugging and incident response times (**hours to minutes**), and aiding cross-functional product architecture improvements.
- Led backend development in **Haskell**, **Groovy** and **Java** for new features and integrations, while **mentoring** junior hires by assigning tasks, guiding and onboarding them on system design; migrated core logic from JavaScript to Haskell, **reducing costs** by 95%
- Designed and maintained a **microservice** based distributed system on **AWS-EKS (Kubernetes)** to orchestrate **Docker** containers, RDS and ElastiCache for storage and cache, to process **100M+ daily transactions** using fault-tolerance patterns and load balancing strategies to ensure 99.999% uptime.

Indian Institute of Technology, Bombay | *Research Intern*

05/2020 - 06/2020

- Researched and implemented deep learning techniques for 3D **point cloud** segmentation using TensorFlow, enhancing accuracy and efficiency in complex geometric analyses.

Publications

A. N. Krishna, Y. L. Chaitra, Atul M. Bharadwaj, K. T. Abbas, **Allen Abraham**, Anirudh S. Prasad, "Real-time Machine Vision System for the Visually Impaired", Springer Nature, (DOI: [10.1007/s42979-024-02741-4](https://doi.org/10.1007/s42979-024-02741-4))

Projects

eYantra Robotics Competition (by Indian Institute of Technology, Bombay) | *Winner and Team Leader*

04/2020

- Led** our robotics team to **1st place** among 1,050 teams in a national competition, integrating real-time systems, maze traversal algorithms, image recognition, robotics design, electronics, and machine learning.

Real time video recommender and analyzer | *Lead Programmer*

04/2024

- Led a three-person team to design and implement a scalable Big Data pipeline using **Python**, Apache **Kafka** and **PySpark** for real-time streaming and large-scale analysis; created **Dockerized** environments and data ingestion scripts to streamline deployment and efficiently process large datasets.

Save For Later

- Created** and **maintained** an **open-source** Chrome extension ([Save for Later](#)), reaching **15,000+ total installations** and sustaining a 4.8-star user rating

Skills

Languages: *Python, C++, Go, Haskell, Java, Groovy, JavaScript*

Technologies: *Git, Linux, SQL, Redis, REST API, TensorFlow, AWS, Docker, Microservices, Object-Oriented Programming*

Software Development: *Agile methodology, Automated Testing, Scalable System Design, Real-time Systems*